Interactive Imitation Learning

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CS 1840 Fall 2026

Announcements

HW4 is out!

HW5 is a mini-project in groups of 2–4.
For HW4, submit your group members and proposed project ideas.

Check the course website for Office Hours:
https://harvard-cs-1840.github.io/2026/

Finishing Up From Last Lecture



1. Offline Imitation Learning: Behavior Cloning

2. Performance difference lemma and its application to proving BC's bound

Performance Difference Lemma

Given two policies $\pi : S \mapsto \Delta(A)$, $\pi' : S \mapsto \Delta(A)$, recall $V^{\pi}(s_0) = \mathbb{E}\left[\sum_{h=0}^{\infty} \gamma^h r(s_h, a_h) \,|\, \pi\right]$

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Performance Difference Lemma (PDL):

$$V^{\pi}(s_0) - V^{\pi'}(s_0) = \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\pi}_{s_0}}\left[\mathbb{E}_{a\sim\pi(\cdot|s)} Q^{\pi'}(s,a) - V^{\pi'}(s)\right]$$

$$:= \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\pi}_{s_0}}\left[\mathbb{E}_{a\sim\pi(\cdot|s)} A^{\pi'}(s,a)\right]$$

$\mathbb{E}_{a\sim\pi}[A^{\pi}(s,a)] = 0$



Analysis of BC

Theorem [BC Performance] With probability at least $1 - \delta$, BC returns a policy $\hat{\pi}$:

$$V^{\pi^\star} - V^{\hat{\pi}} \leq \frac{2}{(1-\gamma)^2}\epsilon$$

↑ SL error

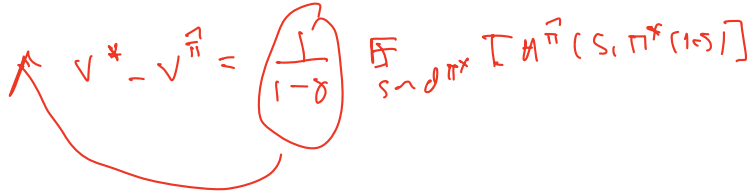
$$\mathcal{D} = \{(s_i^\star, a_i^\star)\}_{i=1}^{N}$$

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$$= \mathbb{E}_{s\sim d^{\pi^\star}} A^{\hat{\pi}}(s, \pi^\star(s)) - \mathbb{E}_{s\sim d^{\pi^\star}} A^{\hat{\pi}}(s, \hat{\pi}(s))$$

$= 0$

$\mathbb{E}_{a\sim\hat{\pi}}[A^{\hat{\pi}}(s,a)] = 0$

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$$= \mathbb{E}_{s\sim d^{\pi^\star}} A^{\hat{\pi}}(s, \pi^\star(s)) - \mathbb{E}_{s\sim d^{\pi^\star}} A^{\hat{\pi}}(s, \hat{\pi}(s)) \quad = \begin{cases} 0 & \text{if } \hat{\pi} = \pi^* \\ \leq \frac{2}{1-\gamma} & \text{if } \hat{\pi} \neq \pi^* \end{cases}$$

$$\leq \mathbb{E}_{s\sim d^{\pi^\star}} \frac{2}{1-\gamma} \mathbf{1}\left\{\hat{\pi}(s) \neq \pi^\star(s)\right\}$$

↑ classification error

$A^\pi \in \left[\frac{-1}{1-\gamma}, \frac{1}{1-\gamma}\right]$

why?

Analysis of BC

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$$\leq \frac{2}{1-\gamma}\epsilon$$

$(1-\gamma)\left[V^* - V^{\hat{\pi}}\right] \leq \frac{2}{1-\gamma}\cdot\epsilon$

$\leq \frac{2}{(1-\gamma)^2}\cdot\epsilon$

Analysis of BC

Theorem [BC Performance] With probability at least $1-\delta$, BC returns a policy $\widehat{\pi}$:

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$$\leq \frac{2}{1-\gamma}\epsilon$$

The quadratic amplification is annoying;
Related to pre-trained LLM hallucination;
Will see how to fix it next lecture

Recap

The Behavior Cloning algorithm:

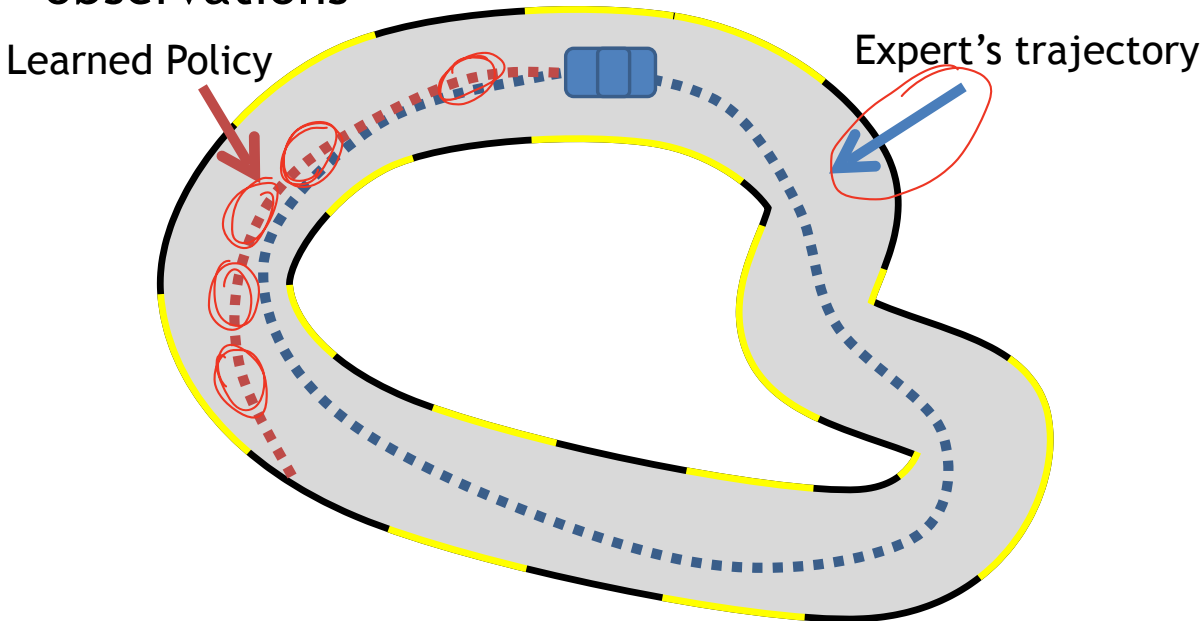
Choose regression (for continuous action) or classification loss $\ell(\pi(s), a)$, and perform SL:

$$\widehat{\pi} = \min_{\pi \in \Pi} \sum_{i=1}^{M} \ell(\pi(s_i^\star), a_i^\star)$$

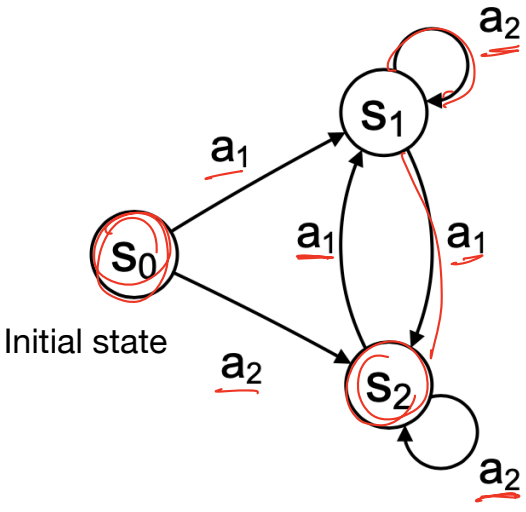
What could go wrong?

[Pomerleau89,Daume09]

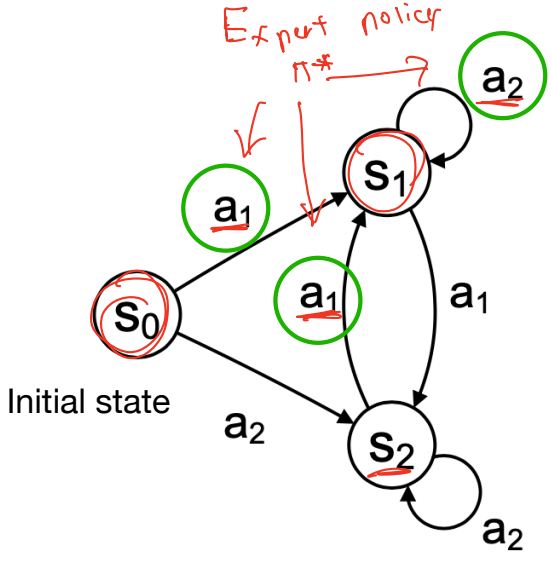
- Predictions affect future inputs/ observations



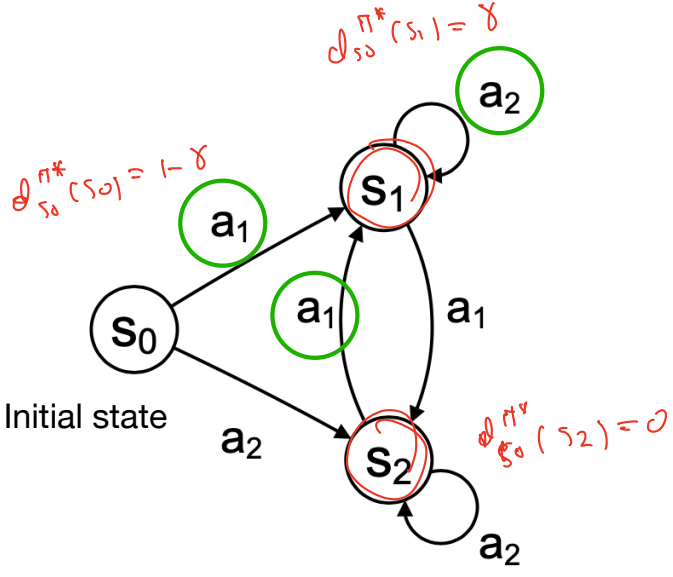
Distribution Shift: Example



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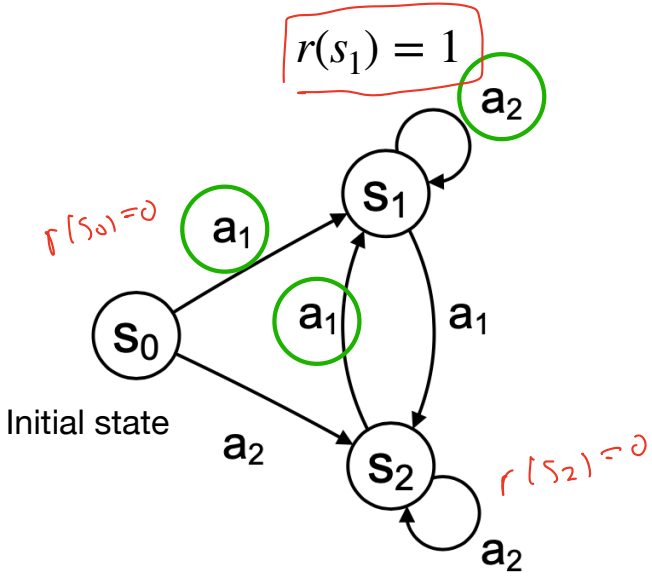


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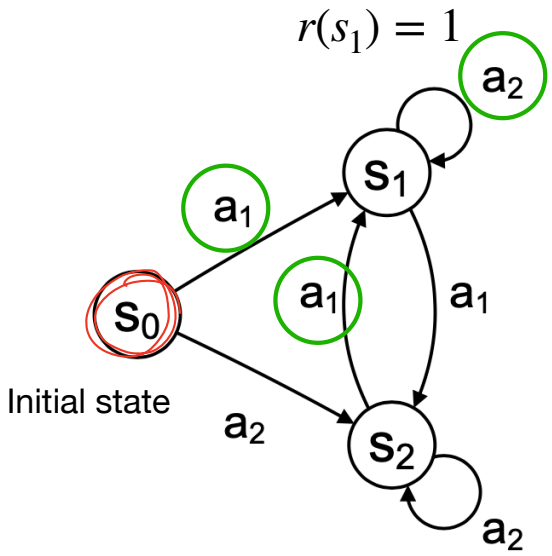
$d_{s_0}^{\pi^\star}(s_0) = 1 - \gamma,\ d_{s_0}^{\pi^\star}(s_1) = \gamma,\ d_{s_0}^{\pi^\star}(s_2) = 0$

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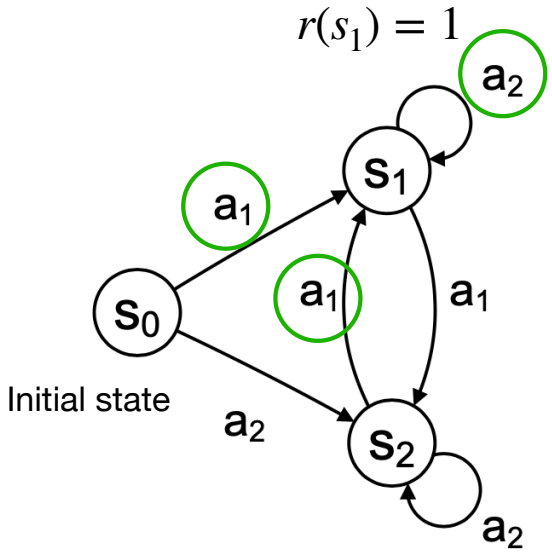
Distribution Shift: Example



$d^{\pi^\star}_{s_0}(s_0) = 1 - \gamma,\ d^{\pi^\star}_{s_0}(s_1) = \gamma,\ d^{\pi^\star}_{s_0}(s_2) = 0$

$V^{\pi^\star}_{s_0} = \dfrac{\gamma}{1-\gamma}$ = $0 + 1\cdot\gamma + 1\cdot\gamma^2 + 1\cdot\gamma^3 + \cdots = \gamma\left[1 + \gamma + \gamma^2 + \cdots\right] = \dfrac{\gamma}{1-\gamma}$

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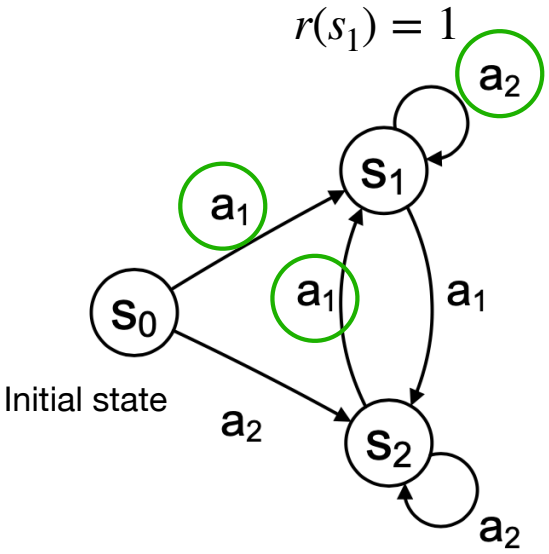
$$V^{\pi^\star}_{s_0} = \frac{\gamma}{1-\gamma}$$

$\epsilon \to$ small

Assume SL returned such policy $\widehat{\pi}$

$$\widehat{\pi}(s_0) = \begin{cases} a_1 & \text{w/ prob } 1 - \epsilon/(1-\gamma) \\ a_2 & \text{w/ prob } \epsilon/(1-\gamma) \end{cases}, \quad \widehat{\pi}(s_1) = a_2,\ \widehat{\pi}(s_2) = a_2$$

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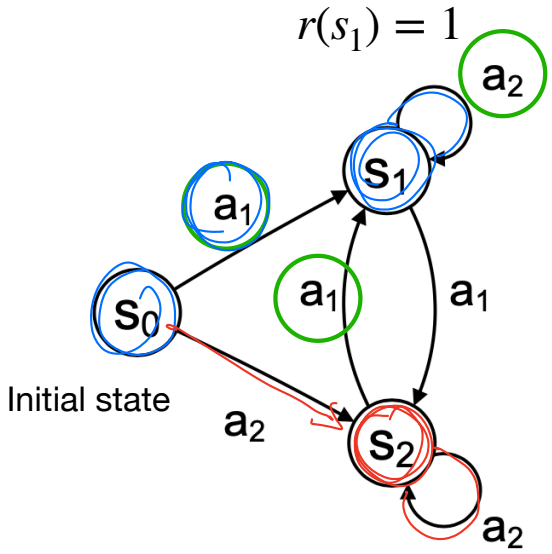
We will have good supervised learning error:

$$\mathbb{E}_{s\sim d^{\pi^\star}_{s_0}} \mathbb{E}_{a\sim\widehat{\pi}(\cdot|s)} \mathbf{1}\left(a \neq \pi^\star(s)\right) = \epsilon$$

At s_0: $d^{\pi^\star}_{s_0}(s_0) \cdot \left[\frac{\epsilon}{1-\gamma}\right] + d^{\pi^\star}_{s_0}(s_1) \cdot 0 + d^{\pi^\star}_{s_0}(s_2) \cdot 1$

$= 1-\gamma \cdot \left(\frac{\epsilon}{1-\gamma}\right) + \gamma \cdot 0 + 0 \cdot 1$

Distribution Shift: Example



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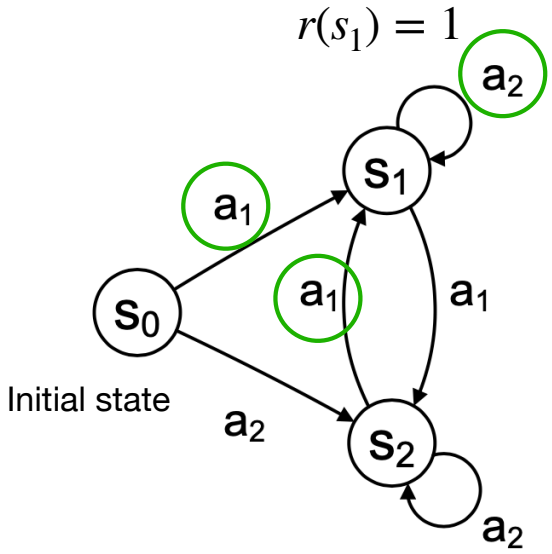
$$\mathbb{E}_{s\sim d^{\pi^\star}_{s_0}} \mathbb{E}_{a\sim\widehat{\pi}(\cdot|s)} \mathbf{1}\left(a \neq \pi^\star(s)\right) = \epsilon$$

But we have quadratic error in performance:

$$V^{\widehat{\pi}}_{s_0} = \frac{\gamma}{1-\gamma} - \frac{\epsilon\gamma}{(1-\gamma)^2} = V^{\pi^\star}_{s_0} - \frac{\epsilon\gamma}{(1-\gamma)^2}$$

$V^{\pi}(s_0) = \left(1 - \frac{a}{(1-\gamma)}\right)\left(\frac{\gamma}{1-\gamma}\right)$ | $V^{\pi}_{a_1} = \left(\frac{a}{1-\gamma}\right)$; $V^{\pi}(s_0) = V^{\pi}_{a_1}(s_0)$

Distribution Shift: Example



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Issue: once we make a mistake at s_0, we end up in s_2 which is not in the training data!

An Autonomous Land Vehicle

In A Neural Network [Pomerleau, NIPS ‘88]



“If the network is not presented with sufficient variability in its training exemplars to cover the conditions it is likely to encounter…[it] will perform poorly”

Question for today:

How to mitigate the distribution shift issue?

Solution:

Interactive Imitation Learning Setting

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Key assumption:
we can query expert $\pi^\star$ at any time and any state during training

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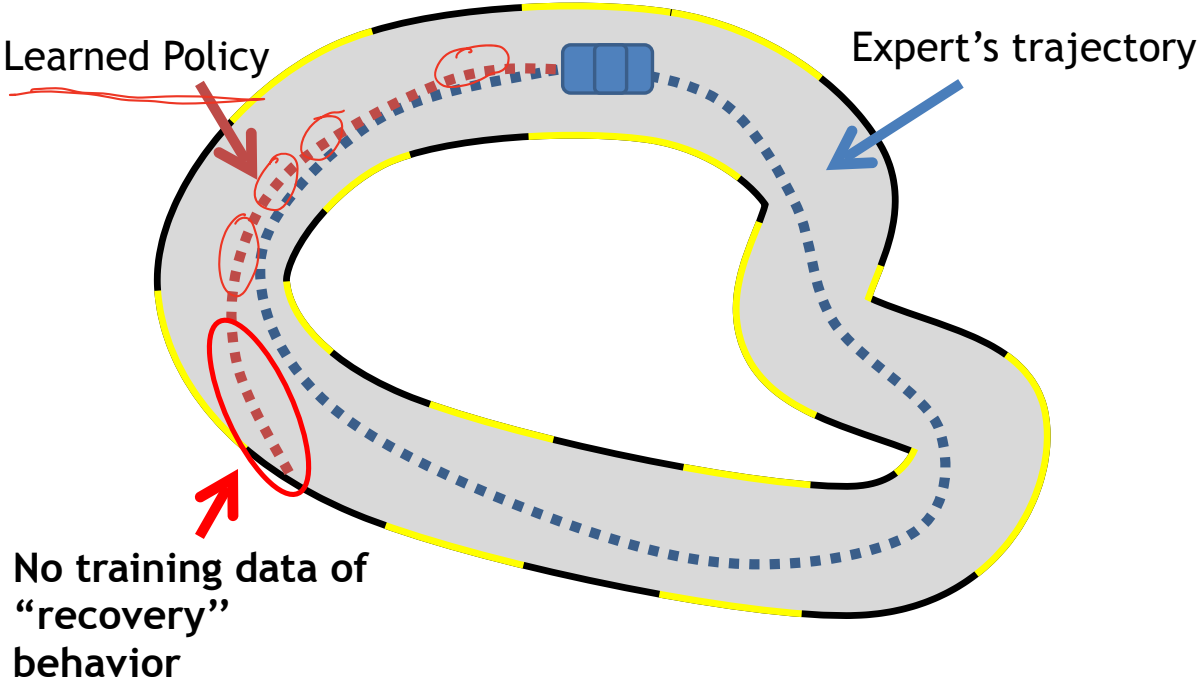
(Recall that previously we only had an offline dataset $\mathscr{D} = (s_i^\star, a_i^\star)_{i=1}^M \sim d_\mu^{\pi^\star}$)

Outline:

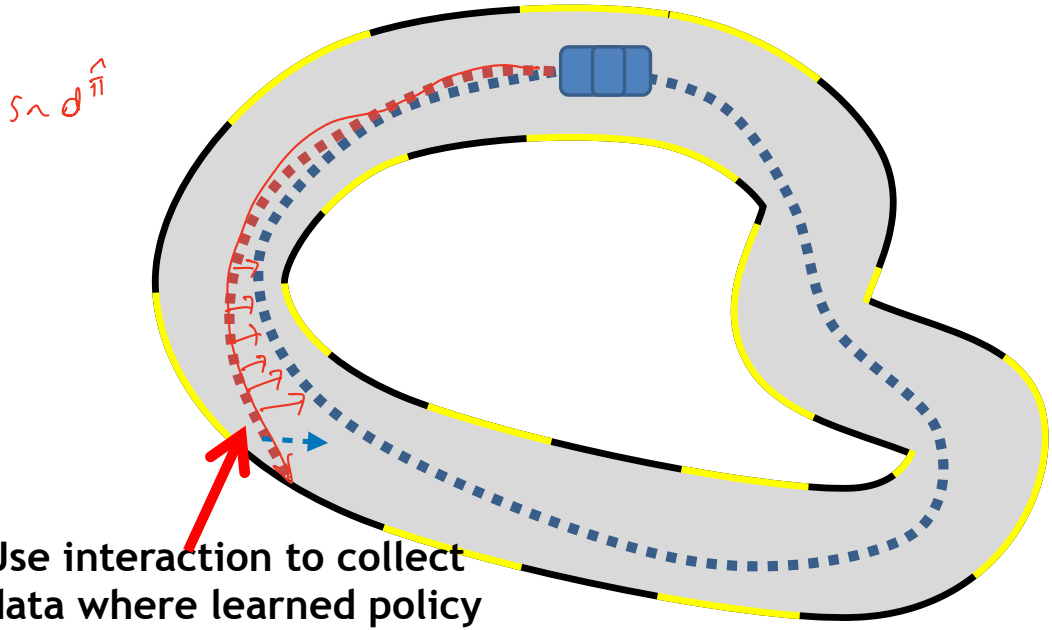
1. The DAgger (Data Aggregation) Algorithm

2. Analysis of DAgger: DAgger as online learning

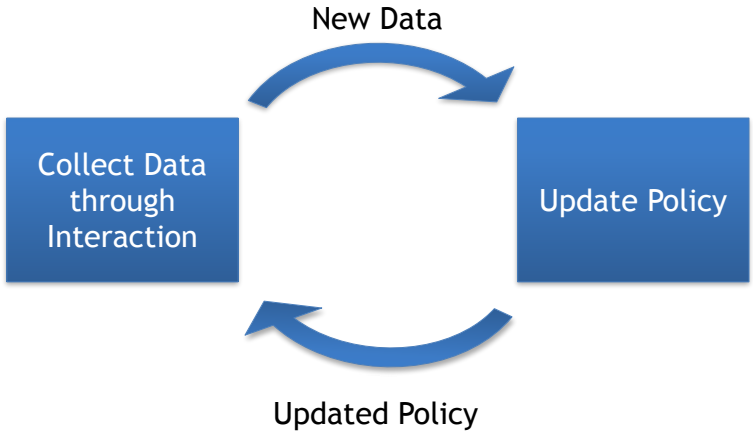
Recall the Main Problem from Behavior Cloning:



Intuitive solution: Interaction



General Idea: Iterative Interactive Approach

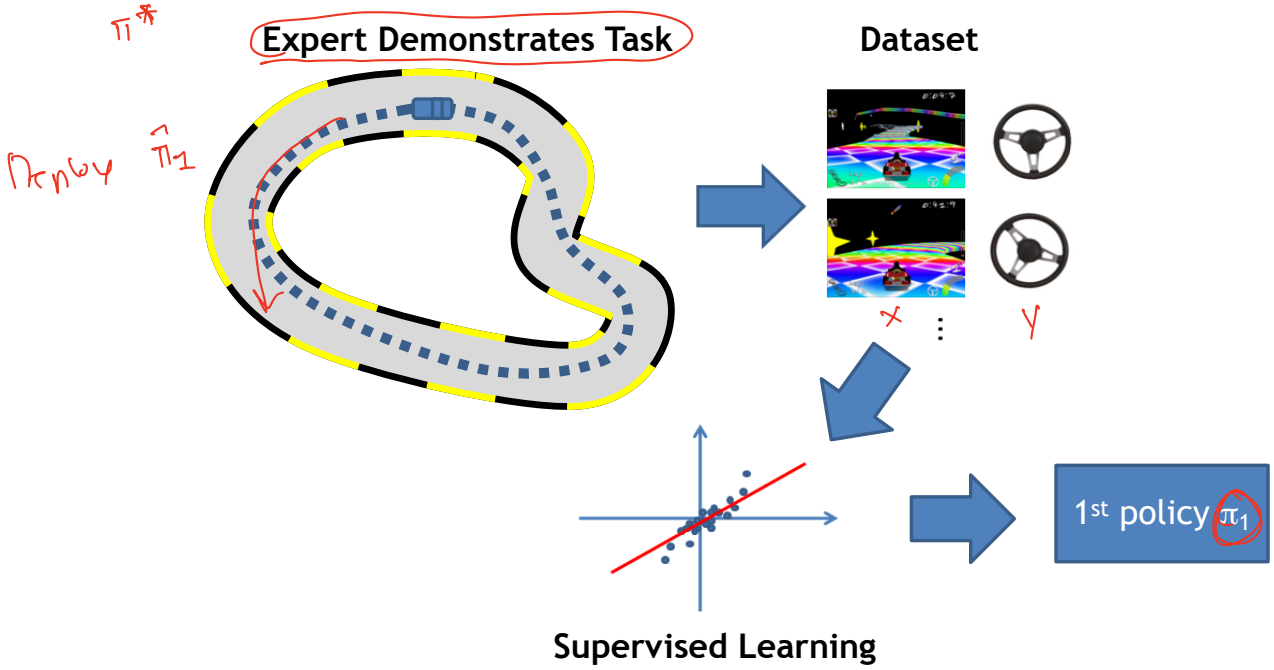


All DAgger slides credit: Drew Bagnell, Stephane Ross, Arun Venktraman

[Ross11a]

DAgger: Dataset Aggregation

0th iteration

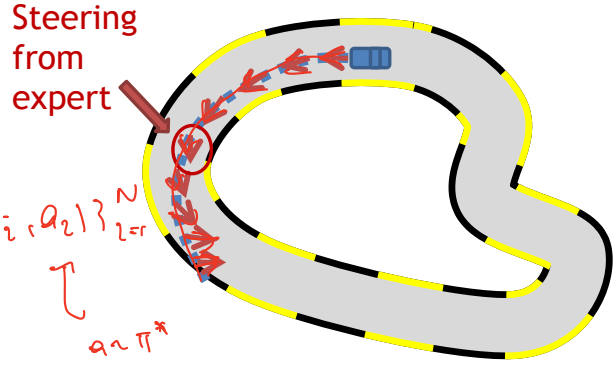


[Ross11a]

DAgger: Dataset Aggregation

1st iteration

Execute π_1 and Query Expert



$D = \{(s_i, a_i)\}_{i=1}^{N}$

$s \sim d^{\pi_1}$

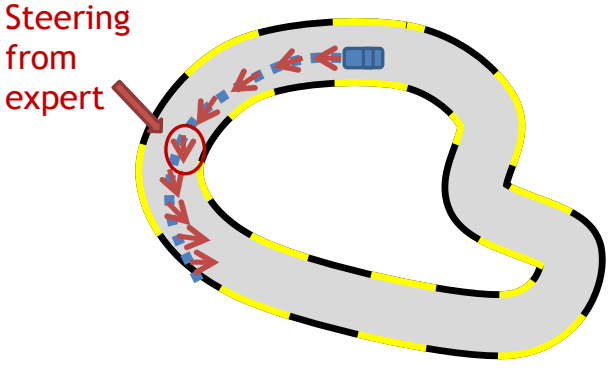
$a \sim \pi^*$

[Ross11a]

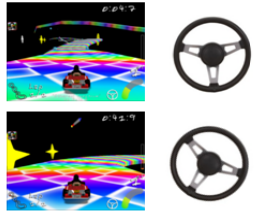
DAgger: Dataset Aggregation

1st iteration

Execute π_1 and Query Expert



New Data



$s \sim d^{\pi_1}$

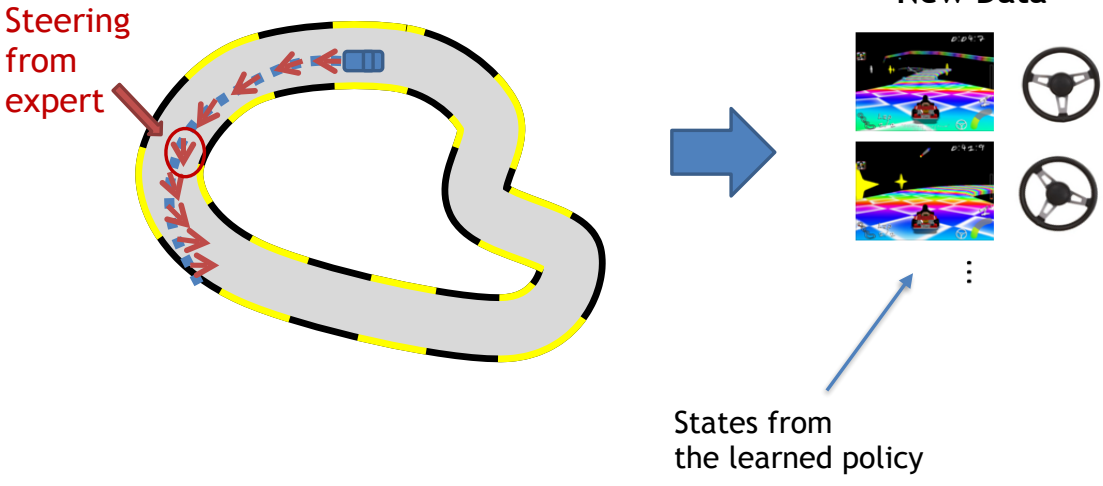
$a \sim \pi^*(\cdot|s)$

[Ross11a]

DAgger: Dataset Aggregation

1st iteration

Execute π_1 and Query Expert

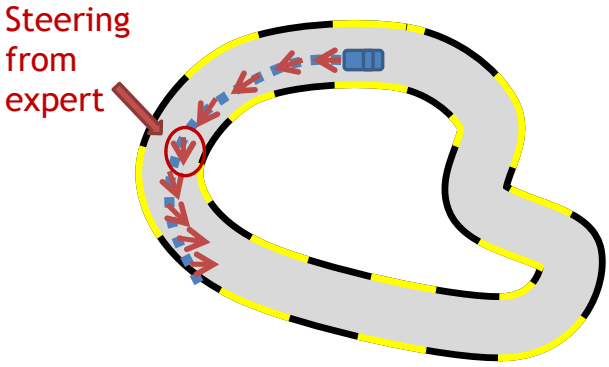


[Ross11a]

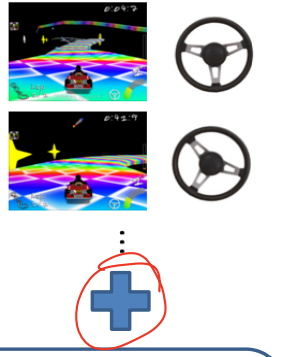
DAgger: Dataset Aggregation

1st iteration

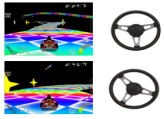
Execute π_1 and Query Expert



New Data



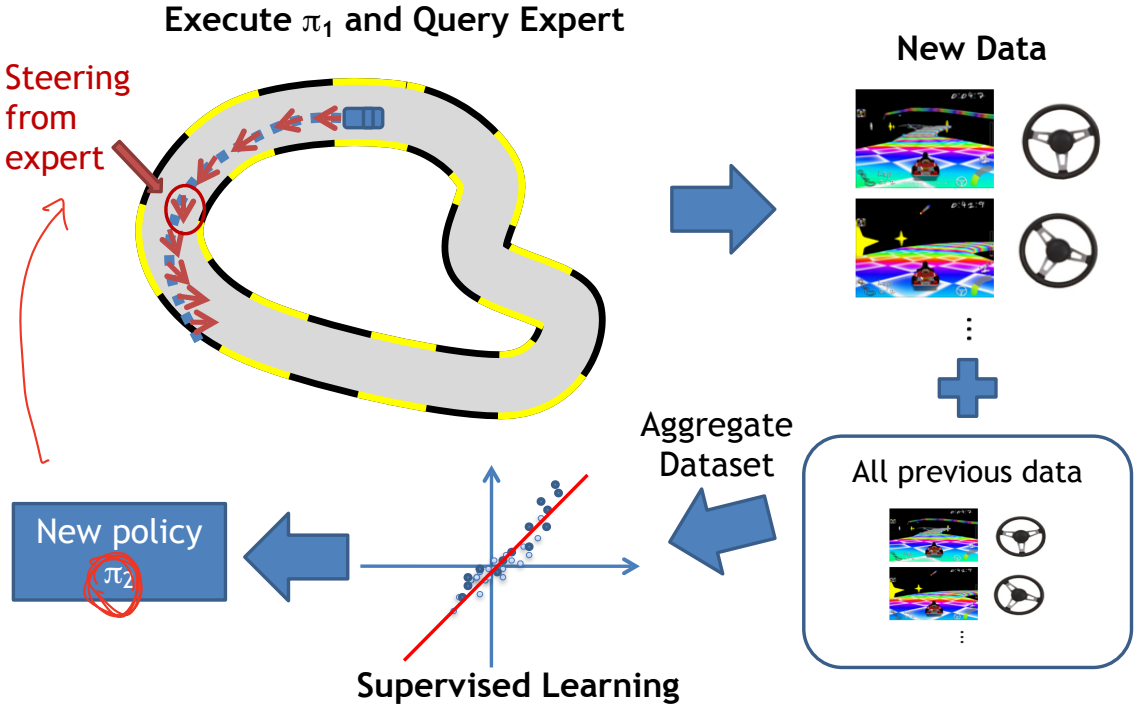
All previous data



[Ross11a]

DAgger: Dataset Aggregation

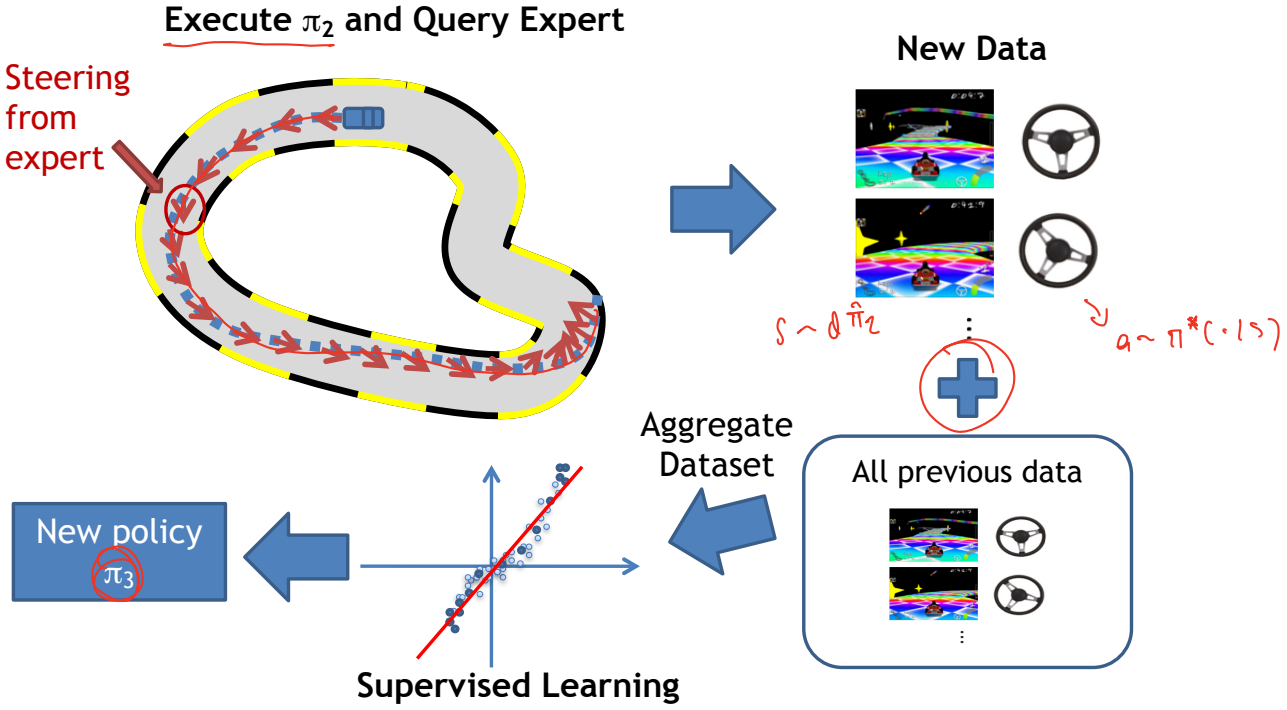
1st iteration



[Ross11a]

DAgger: Dataset Aggregation

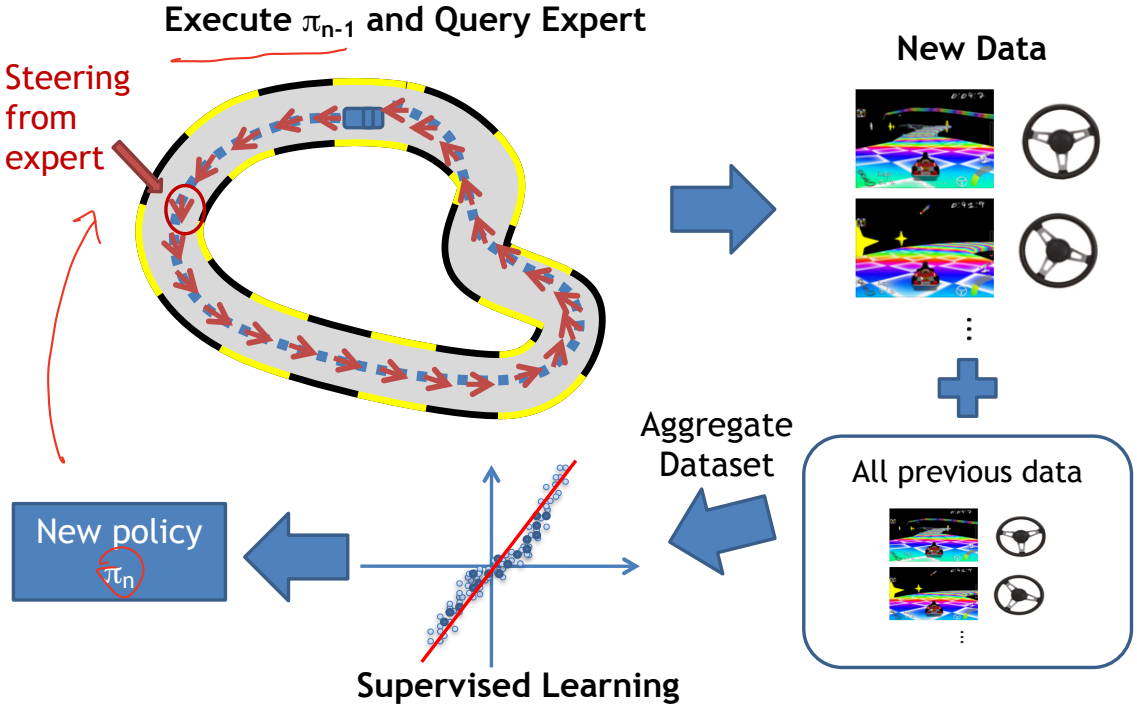
2nd iteration



[Ross11a]

DAgger: Dataset Aggregation

nth iteration



Success!

[Ross AISTATS 2011]



Success!

[Ross AISTATS 2011]



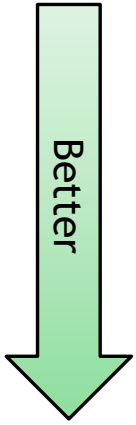
Success!

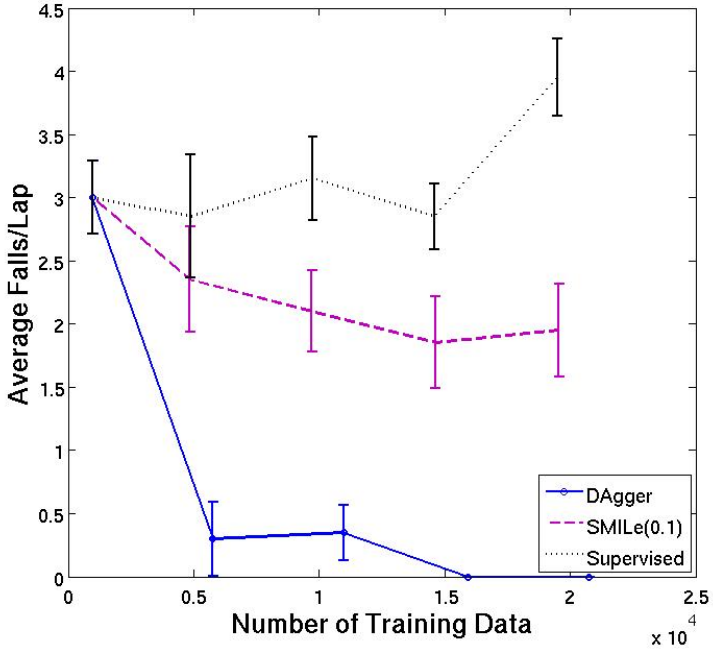
[Ross AISTATS 2011]



[Ross AISTATS 2011]

Average Falls/Lap






FPS: 24
Attempt: 1 of 1
AgentLinear
Selected Actions:
RIGHT
SPEED


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More fun than Video Games…



[Ross ICRA 2013]

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Forms of the Interactive Experts

Interactive Expert is expensive, especially when the expert is human…

But expert does not have to be human…

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Example: high-speed off-road driving
[Pan et al, RSS 18, Best System Paper]



Fig. 4: The AutoRally car and the test track.

Forms of the Interactive Experts

Interactive Expert is expensive, especially when the expert is human…

But expert does not have to be human…

Example: high-speed off-road driving
[Pan et al, RSS 18, Best System Paper]



Fig. 4: The AutoRally car and the test track.

Goal: learn a racing control policy that maps from data on cheap on-board sensors (raw-pixel imagine) to low-level control (steer and throttle)

Forms of the Interactive Experts

Interactive Expert is expensive, especially when the expert is human…

But expert does not have to be human…

Example: high-speed off-road driving
[Pan et al, RSS 18, Best System Paper]



Fig. 4: The AutoRally car and the test track.

Goal: learn a racing control policy that maps from data on cheap on-board sensors (raw-pixel imagine) to low-level control (steer and throttle)



→ Steering + throttle

(a) raw image

Forms of the Interactive Experts

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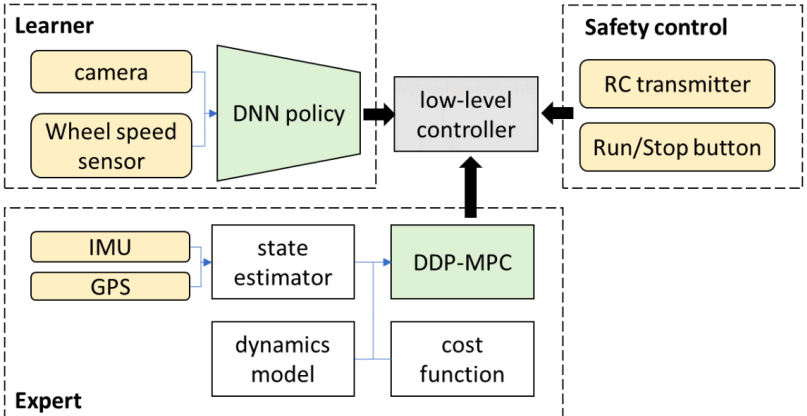
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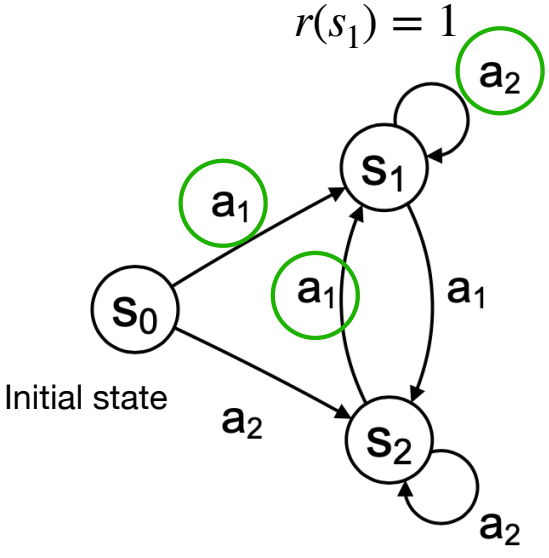
Named Entity Recognition

Input: After completing his Ph.D., Ellis worked at Bell Labs from 1969 to 1972 on probability theory..

π_i O O O O O PER O

π^* O O O O O PER O O ORG ORG O O O O O O O

Distribution Shift: Example



$$d_{s_0}^{\pi^\star}(s_0) = 1 - \gamma,\ d_{s_0}^{\pi^\star}(s_1) = \gamma,\ d_{s_0}^{\pi^\star}(s_2) = 0$$

$$V_{s_0}^{\pi^\star} = \frac{\gamma}{1-\gamma}$$

Assume SL returned such policy $\widehat{\pi}$

$$\widehat{\pi}(s_0) = \begin{cases} a_1 & \text{w/ prob } 1 - \epsilon/(1-\gamma) \\ a_2 & \text{w/ prob } \epsilon/(1-\gamma) \end{cases}, \quad \widehat{\pi}(s_1) = a_2,\ \widehat{\pi}(s_2) = a_2$$

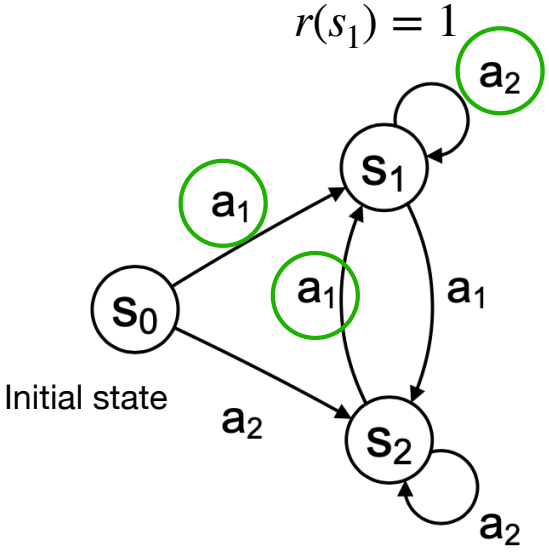
We will have good supervised learning error:

$$\mathbb{E}_{s \sim d_{s_0}^{\pi^\star}} \mathbb{E}_{a \sim \widehat{\pi}(\cdot|s)} \mathbf{1}\left(a \neq \pi^\star(s)\right) = \epsilon$$

But we have quadratic error in performance:

$$V_{s_0}^{\widehat{\pi}} = V_{s_0}^{\pi^\star} - \frac{\epsilon\gamma}{(1-\gamma)^2}$$

Distribution Shift: Example



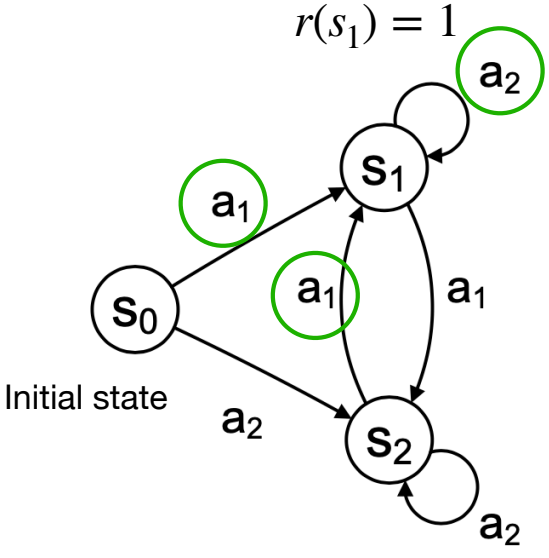
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Why DAgger can fix this problem?

Outline:

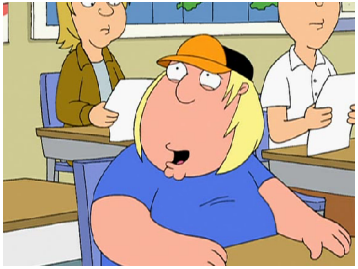


1. The DAgger (Data Aggregation) Algorithm

2. Analysis of DAgger: DAgger as online learning

Online Learning

Learner



convex Decision set Θ

Adversary

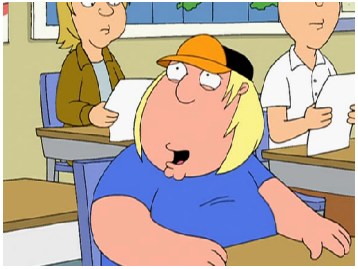


...

[Vovk92,Warmuth94,Freund97,Zinkevich03,Kalai05,Hazan06,Kakade08]

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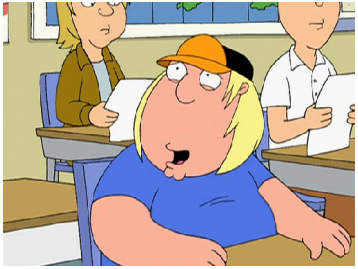


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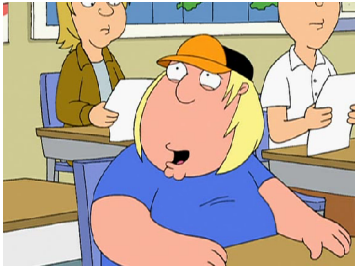
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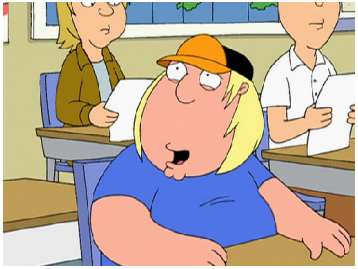
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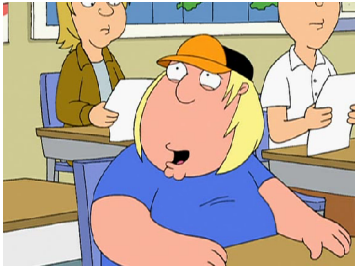
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$$\text{Regret} = \sum_{t=0}^{T-1} \ell_t(\theta_t) - \min_{\theta \in \Theta} \sum_{t=0}^{T-1} \ell_t(\theta)$$

[Vovk92,Warmuth94,Freund97,Zinkevich03,Kalai05,Hazan06,Kakade08]

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Can we perform linear regression online with non i.i.d (or even adversary) data?

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Adversary seems too powerful…

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$$\sum_{i=0}^{T-1} \ell_i(\theta_i) - \min_{\theta \in \text{Ball}} \sum_{i=0}^{T-1} \ell_i(\theta) = O\left(\sqrt{T}\right)$$

Generally, Follow-the-Regularized-Leader is no-regret

At time step t, learner has seen $\ell_0, \ldots \ell_{t-1}$, which new decision she could pick?

$$\textbf{FTRL: } \theta_t = \min_{\theta \in \Theta} \sum_{i=0}^{t-1} \ell_i(\theta) + \lambda R(\theta)$$

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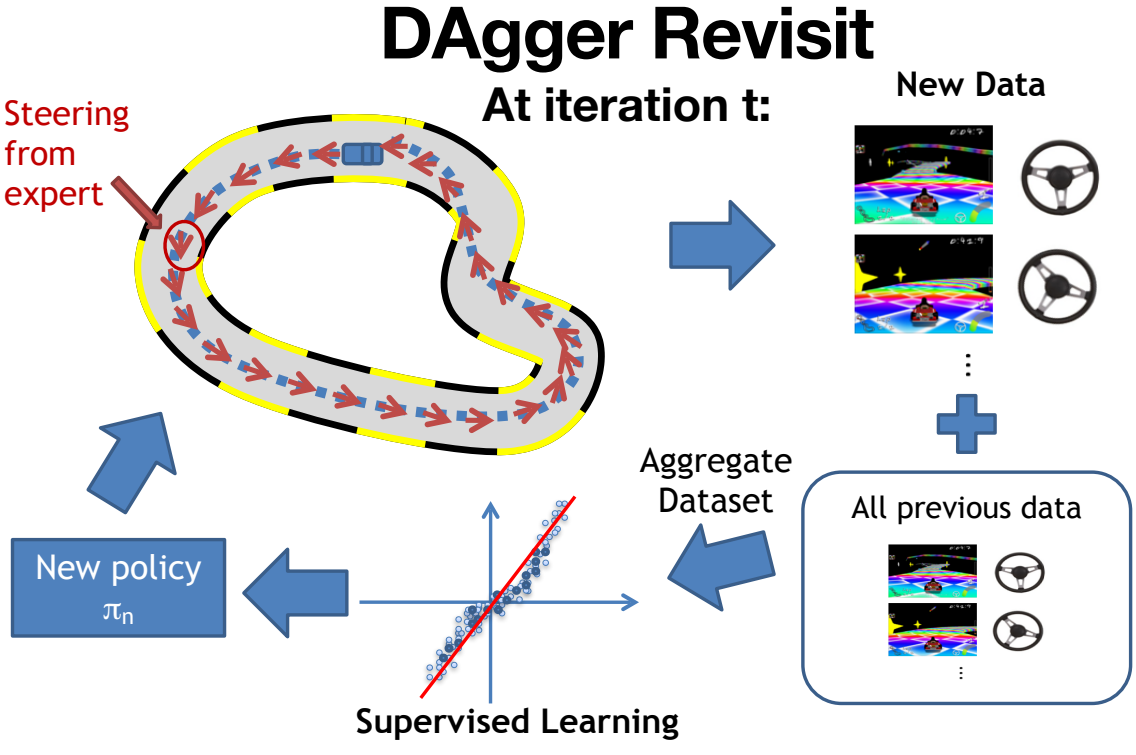
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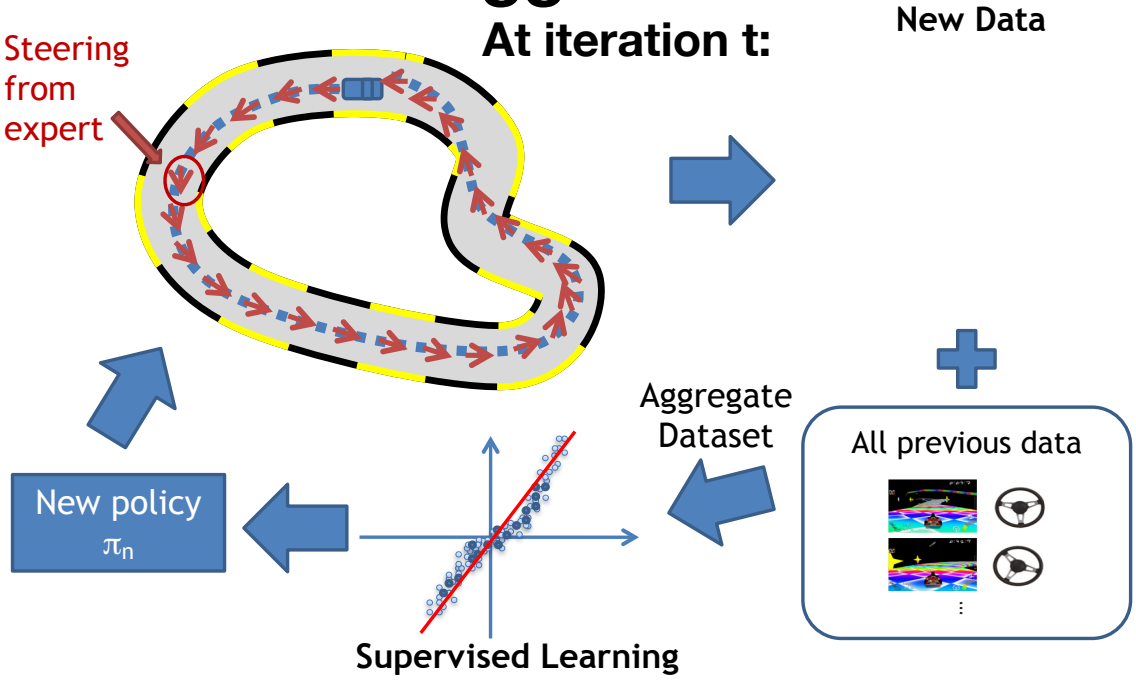
Informal Theorem (FTRL): when things are convex, FTRL is no-regret, i.e.,

$$\frac{1}{T}\left[\sum_{t=0}^{T-1} \ell_t(\theta_t) - \min_{\theta \in \Theta} \sum_{t=0}^{T-1} \ell_t(\theta)\right] = O\left(1/\sqrt{T}\right)$$

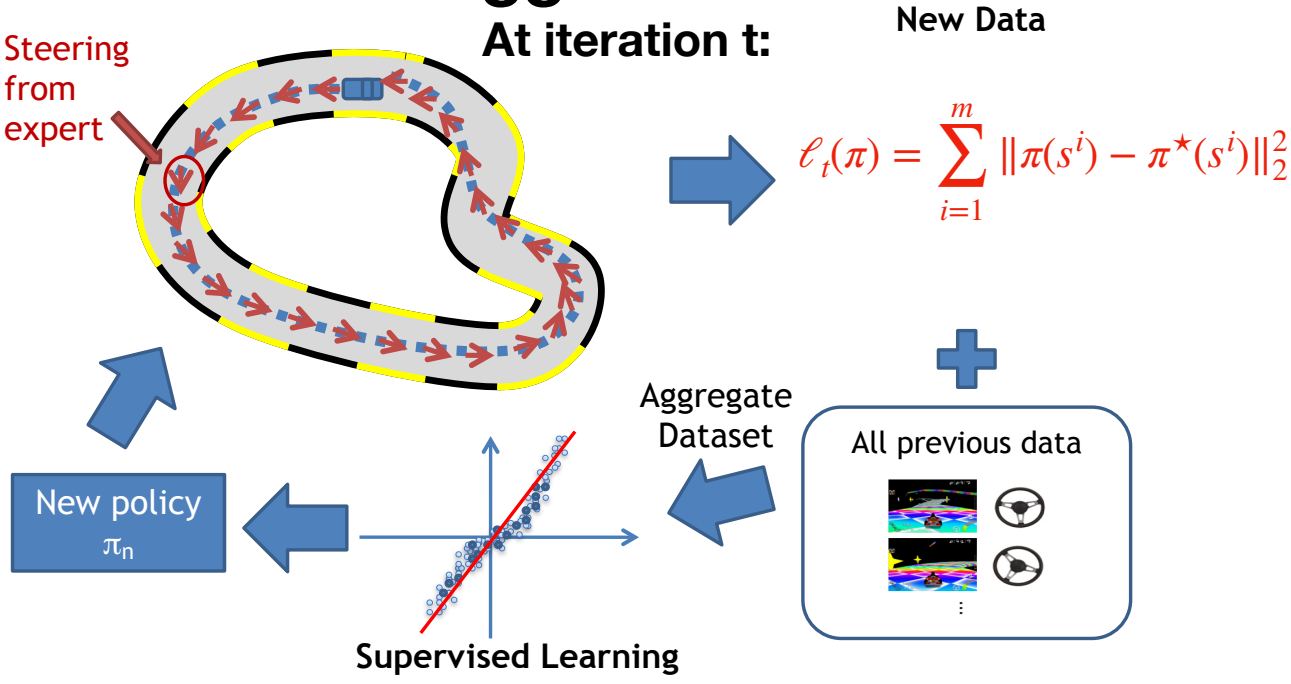
DAgger Revisit



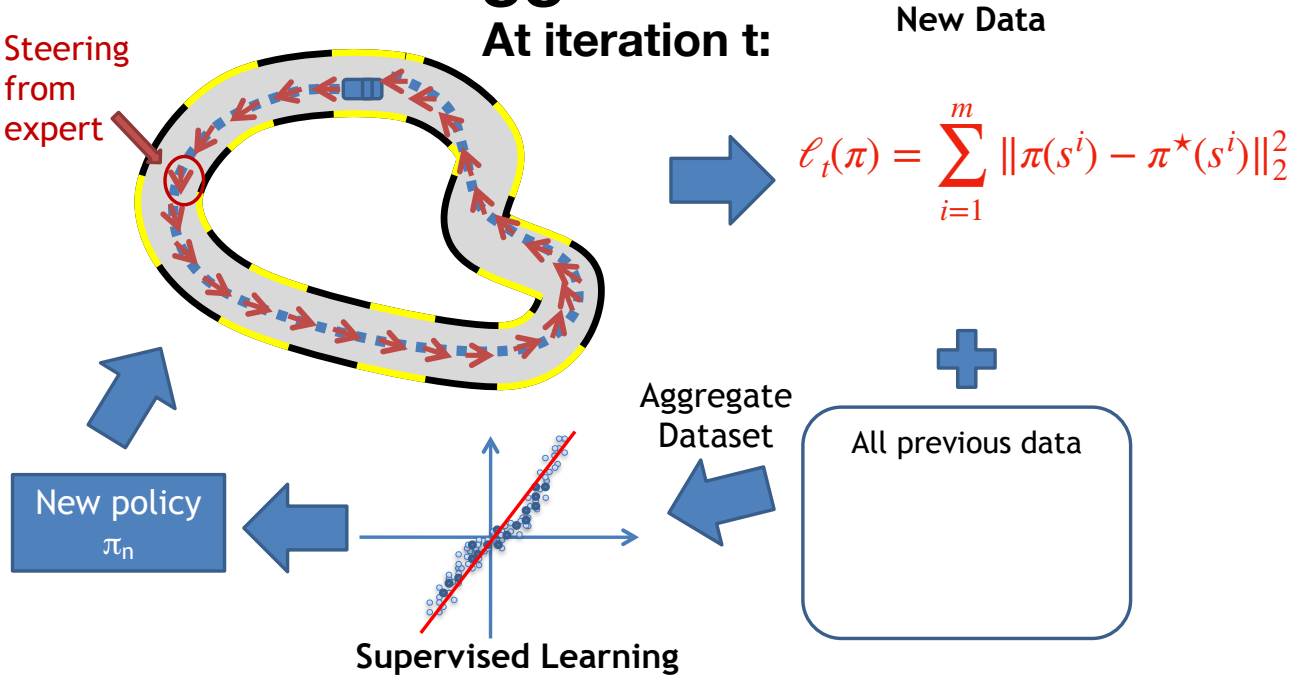
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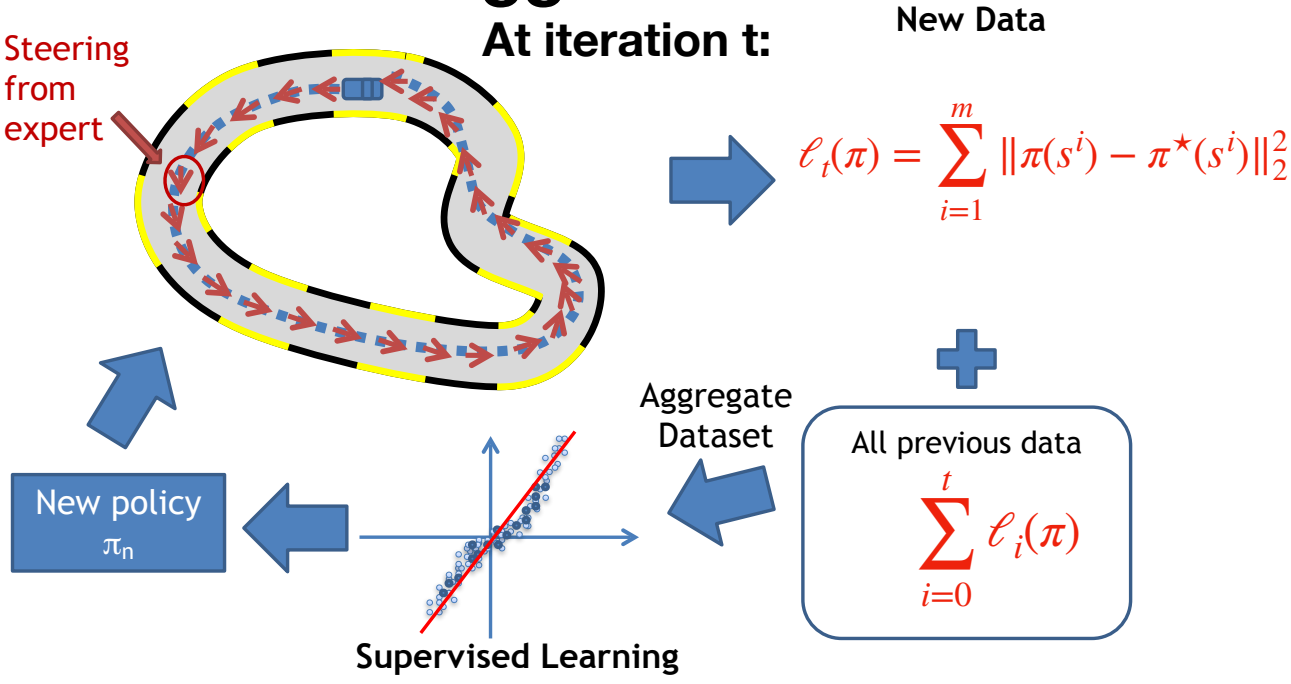
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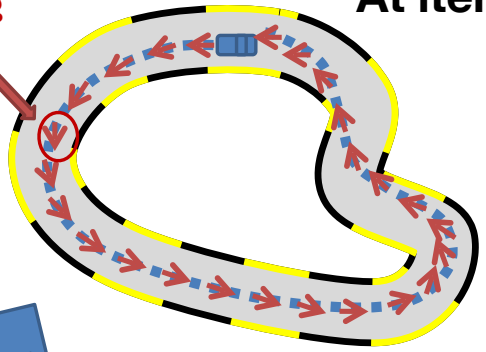
DAgger Revisit



DAgger Revisit

At iteration t:

Steering from expert



New Data

$$\ell_t(\pi) = \sum_{i=1}^{m} \|\pi(s^i) - \pi^\star(s^i)\|_2^2$$

Aggregate Dataset

All previous data

$$\sum_{i=0}^{t} \ell_i(\pi)$$

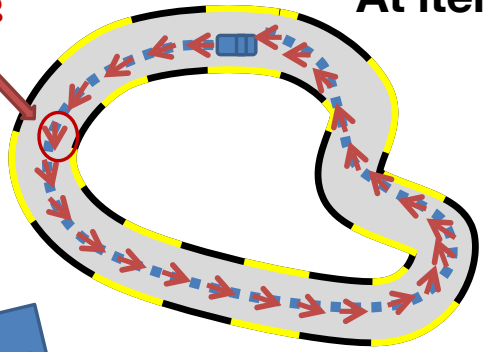
New policy π_n

Supervised Learning

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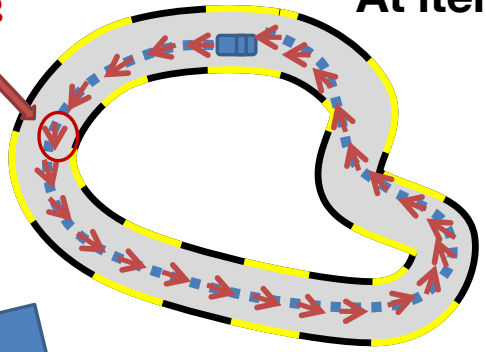
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Data Aggregation = Follow-the-Regularized-Leader Online Learner

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For $t = 0 \to T - 1$:

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3. **Update policy via Supervised-Learning**: $\pi^{t+1} = \text{SL}\left(\mathscr{D}\right)$

2. We can see that DAgger is essentially an online-learning algorithm (FTRL)

Feedback and Attendance

https://bit.ly/4gYn2nC

